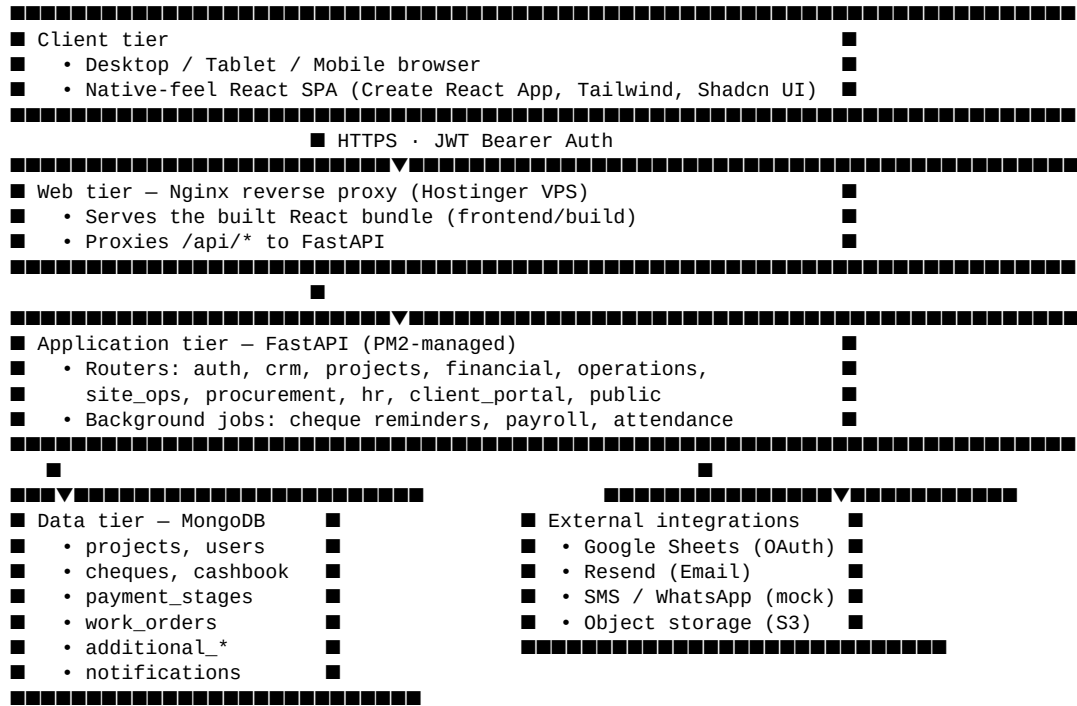


MyHomeUSB Construction CRM — Full System Architecture

Last updated: Feb 2026

1. Tech Stack & Topology



Key runtime conventions

- Single VPS deploy via `sshpass-driven git pull && yarn build && pm2 restart backend`.
- All backend routes are mounted under `/api` (Nginx forwards exactly that prefix to FastAPI).
- All URLs, ports, MongoDB credentials live in `.env` files — never hard-coded.
- Auto-redirect to `/client-portal/{projectId}` for single-project clients (Feb 2026).

2. Data Model Cheat-Sheet

Collection	Purpose
`users`	Every login + role + profile + assignment data
`leads`	Pre-sales / sales pipeline rows
`projects`	Master project ledger (status, team, location, contract data)
`payment_stages`	CRE-side schedule of client payments per project
`project_work_orders`	Contractor/labour WO with `scope_items[]`, `stages[]`, `additional_sections[]`, `additional_work[]`
`material_requests` / `_receipts`	SE flow → Procurement
`labour_expenses`	DLR + payment_request ladder
`cheques`	Cheque Management (Received → Opened → Issued → Bounced)
`cashbook`	Single source of truth for every released payment & received income

Collection	Purpose
`addition_sections`	Per-WO grouping for Additional work (Claimable / Non-Claimable / Rework-SE / Rework-Client)
`notifications`	Per-user in-app notifications
`petty_cash`	PM/SE petty cash requests + issuance
`attendance` / `payroll`	HR module (under build)

3. Users & Interfaces — Who Logs In Where

The system has **23 distinct roles**, each with a tailored dashboard.

#	Role	Login UI	Primary Workspace
1	`super_admin`	Web	Super Admin Dashboard (all modules unlocked)
2	`general_manager`	Web	GM Approvals + Read-only org view
3	`cre` (Client Relationship Officer)	Web	CRE Dashboard → Project payment ladder
4	`accountant`	Web	Finance Board (Cashbook · Approvals · Cheque Mgmt · Income)
5	`project_manager`	Web	PM Dashboard (Projects · Team · Material · Labour · Petty Cash)
6	`associate_pm`	Web	Sr. SE board + limited PM actions
7	`sr_site_engineer`	Web	SE board with elevated role permissions
8	`planning`	Web	Planning workspace + Project Detail editor
9	`planning_person`	Web	Project-scoped Planning view
10	`procurement`	Web	Procurement Board (All Projects · Vendors · Material Stock · Vendor Book)
11	`site_engineer`	Web	SE Dashboard (Projects · Work Orders · DLR · Material · Petty Cash)
12	`vendor`	Web	Vendor Portal (POs · Material Dispatch)
13	`client`	Web (Client Portal)	Single-project dashboard (Payments · Visits · Cashflow)
14	`pre_sales`	Web	CRM Pre-Sales Pipeline
15	`sales`	Web	CRM Sales Pipeline
16	`marketing_head`	Web	Marketing analytics + Lead distribution config
17	`architect`	Web	Architect workspace (drawings, design packages)
18	`super_architect`	Web	Architect + Approvals
19	`hr`	Web	HR module (Attendance / Leave / Payroll)
20	`quality_check`	Web	QC inbox (RAB QC approvals + site visits)
21	`prospect`	Mobile-first	Quote/Package viewer (read-only)

Plus background "users":

- **Public** (no login) — landing site, package showcase, lead capture form.
- **System** — cron-driven entries (cheque reminders, escrow sweeps).

4. Per-Role Function Inventory

Below is what each persona does day-to-day (function = a discrete capability exposed on their dashboard).

4.1 `super\_admin`

Has every function of every role. Plus exclusive:

- User CRUD, role re-assignment, slot management.
- Org-wide reports, audit log, cashbook reversal.
- Master config: payment templates, WO templates, stage templates, GP/IS %.

4.2 `general\_manager`

1. View Org KPIs (projects · revenue · margin).
2. Approve project advancement from Planning.
3. Approve high-value RAB releases (above threshold).
4. View all cashbook entries (read-only).
5. Org-wide notification inbox.

4.3 `cre`

1. Create new project from a converted Sales lead.
2. Set up the client-side payment_stages ladder.
3. Verify first cheque + raise to Accountant for verification.
4. Track project closure (% paid · pending visits).
5. Push project to Planning when verification clears.
6. Maintain client roster · upload contract docs.

4.4 `accountant`

1. **Cashbook** — debit / credit ledger (all expense & income).
2. **Approvals** sub-tabs:
 - Income (CRE-raised cheques)
 - Material Payments (Procurement-raised POs)
 - Labour Payments (SE-raised RABs · _multi-stage releases in one click_)
 - Petty Cash (PM/SE issuance)
3. **Cheque Management** — Received · Opened · Awaiting CRE · Issued · Bounced · Disabled (single source of every cheque in the company).
4. **Income** — view received funds + reconcile.
5. **Suspense Ledger** — track contractor credits and apply on next payment.
6. **Reject / Re-route** any of the above with notes back to the originator.

4.5 `project\_manager`

1. **All Projects** — Sr. SE-style table view; clicking a row opens the SE workspace.
2. **Change Stage** (Yet to Start → Sub-structure → Super-structure → Finishing → Handover).
3. **Assign Team** to projects (SE / Sr.SE / PMs).
4. **Petty Cash** sub-tabs:
 - PM Petty Cash (own requests)
 - Approve SE Petty Cash
 - Issued history

5. **Approve Material Requests** raised by SE.
6. **Approve Labour RABs** raised by SE (step 1 of the approval ladder).
7. **Approve Mark Work Complete** for stage closure.
8. Org-wide Notifications.

4.6 `associate\_pm`

- All Sr. SE functions.
- Limited PM approvals (PM petty cash + RAB stage approval).

4.7 `sr\_site\_engineer`

- All SE functions.
- **Submit/Delete RABs** (elevated permission).
- Bulk DLR entries.

4.8 `planning` & `planning\_person`

1. **Project Detail editor** — full read/write on every WO.
2. Add / Edit / Delete **Scope items** (with searchable Unit dropdown).
3. Add / Edit / Delete **Payment Stages** (Fixed amount or % of contract).
4. Add / Edit / Delete **Additional Sections** (4 buckets: Claimable / Non-Claimable / Rework-SE / Rework-Client).
5. Per-section lock/unlock toggle (cascades to items).
6. Per-item lock/unlock toggle (gated on section being unlocked).
7. **Approve RAB** (step 3 of the approval ladder: SE → PM → QC → **Planning** → Accountant).
8. **Reject** with notes.
9. Recompute contract totals, freeze WOs.

4.9 `procurement`

1. **All Projects** — material requirement aggregator.
2. **Material Vendors** sub-tab (Vendor view · Materials view).
3. **Material Stock** — receipt vs consumption ledger.
4. **Vendor Book** — Orders · Credits · Summary per vendor.
5. **Material Edit Popup** — Details · Vendors split tabs.
6. Multi-Mode Payment Release (cheque + cash mix).
7. Raise Material PO to Accountant.

4.10 `site\_engineer`

1. **My Projects** picker (assignments-driven).
2. **Work Orders** board (per contractor):
 - **Payment Schedule** (status pills: All / Open / Completed / Locked Stages).
 - **Additional** (4 sub-tabs: Claimable / Non-Claimable / Rework-SE / Rework-Client).
 - **Additional RAB** (history scoped to additional stages).
 - **Total RAB's** (excludes additional stages).

- **DLR** (Daily Labour Report).

3. **Request RAB** dialog:

- Multi-stage allocation with stage-group scoping.

- Per-section expand → per-item Pick checkbox.

- Auto-distribute or manual amounts.

- Section selection scoped by claim_type group.

4. **Edit pending RAB** — same dialog as Request RAB in edit mode.

5. **Mark Work Complete** for a stage.

6. **Material Requests** (raise to PM → Procurement).

7. **Material Receipts** (sign-off on arrival).

8. **Petty Cash** request.

9. **DLR entry** (workers, hours, attendance).

4.11 `vendor`

1. View their POs.

2. Confirm / dispatch.

3. Upload invoices.

4. View payment status.

4.12 `client`

1. **Single-project dashboard** (auto-redirect on login for 1-project clients).

2. View **payment schedule** + status (Paid / Approved / Pending).

3. **Approve / Reject** Claimable Additional charges.

4. **Cashflow** — money in vs out (filtered: hides Non-Claimable + Rework-SE rows).

5. **Site Visits** — view scheduled visits, confirm presence.

6. **Documents** — contract, invoices, completion certificate.

7. Push notifications via in-app banner.

4.13 `pre\_sales`

1. CRM Lead inbox (auto-distributed + solo-mode fallback).

2. First-contact log.

3. Convert qualified lead → push to Sales.

4.14 `sales`

1. Pipeline view (Stages: New · Quoted · Negotiation · Won · Lost).

2. Slot Management (Phase 2 — show slot beside user's name).

3. Convert Won → push to CRE.

4.15 `marketing\_head`

1. Lead-source analytics.

2. Distribution rules editor (round-robin / weighted / solo-mode).

3. Campaign ROI.

4.16 `architect` / `super\_architect`

1. Upload design packages.
2. Version control on drawings.
3. Tag drawings to projects.
4. (Super) Approve revisions.

4.17 `hr`

1. Attendance grid.
2. Leave (CL / SL) approvals.
3. Manual punch adjustments.
4. Payroll generation (Phase 2).

4.18 `quality\_check`

1. **RAB QC inbox** (step 2 of the approval ladder: SE → PM → **QC** → Planning).
2. Site visit reports.
3. Reject with rework reason → creates Rework (SE) additional item.

4.19 `prospect`

1. View shared quote.
2. Browse package gallery.
3. Trigger callback.

5. End-to-End Business Flow (Lead → Handover)



SE raises RAB
 ↓ PM approves
 ↓ QC approves
 ↓ Planning approves
 ↓ Accountant releases (cashbook OUT)
 ■■■ Cash (Petty / Suspense)
 ■■■ Cheque (linked to issued cheque)
 ■■■ Bank transfer (current / savings)

For Additional Work:
 Planning unlocks section + items
 ↓ SE bills section (or specific picked items)
 ↓ Same approval ladder
 ↓ Client Portal hides Non-Claimable + Rework-SE rows

For Multi-Stage Bills:
 SE checks N section/stage boxes → allocates amount
 ↓ One RAB request with stage_breakdown[]
 ↓ Approval ladder runs once
 ↓ Accountant releases as ONE payment, ONE cashbook entry

Stage Closure → PM Mark Work Complete
 ■
 ▼
 Project moves Sub-structure → Super-structure → Finishing → Handover
 ■
 ▼
 CRE closes (final cheque + handover certificate)
 ■
 ▼
 COMPLETED

6. Cross-Cutting Modules

- **Cheque Management** — every cheque in the company surfaces here (Received → Opened → Issued → Bounced / Disabled). Project column populated end-to-end (Feb 2026 fix).
- **Suspense Ledger** — auto-credits when a cheque payment exceeds the bill amount; auto-applies on next payment to the same contractor.
- **Notifications** — per-user inbox + push banner; backed by [db.notifications](#).
- **Audit Trail** — every payment_request carries [pm_approved_by](#), [qc_approved_by](#), [planning_approved_by](#), [accountant_released_by](#), [released_at](#) snapshots.
- **Public Showcase** — public package gallery, no login required, drives leads.
- **Mobile** — fully responsive; prospect role is mobile-first.

7. Approval Ladder Summary

Module	Originator	Step 1	Step 2	Step 3	Step 4 (Release)
Labour RAB	Site Engineer	PM	QC	Planning	Accountant
Material PO	Site Engineer	PM	—	Planning	Accountant
Petty Cash (SE)	Site Engineer	PM	—	—	Accountant
Petty Cash (PM)	PM	—	—	—	Accountant
Income (Cheque)	CRE	Accountant	—	—	(auto-cashbook)
Project Advancement	Planning	GM	—	—	Super Admin
Project Stage	PM	—	—	—	(direct)

End of architecture document.